

MODULE GUIDE 2026/27

Cloud Native Development

Module Code: **COM682** (**17380**)

Programme(s): **BSc (Hons) Computer Science;**
BSc (Hons) BEng CompScience;
BSc (Hons) Software Engineering;
BSc (Hons) PT;

School of Computing
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Notice: Please be aware that the information provided within the module guide is subject to change.

1. Table of Contents

Welcome 3
1. Module Overview and Communication 4
2. Sustainable Development Goals (SDGs) and Graduate Attributes (GAs) in your module 6
Sustainable Development Goals (SDGs) 6
3. Aim and Learning Outcomes 7
Using Gen AI Tools in this Module 7
4. Assessment and Feedback 8
Summary Assessment Information 8
Reasonable Adjustments 8
Coursework 1 9
Coursework 2 12
5. Learning Resources 14
Library's Support Services 14
Blackboard Learning Support 15
IT Support 15
IT Resources: 15
How to Get Help: 15
How to contact IT Service Desk: 15
6. Organisation and Management 16
Teaching and Learning Plan 18
7. Student Voice and Support Services 25
UUSU Advice Bureau Service 25
UU Student Wellbeing Service 25
UU Student Success Centre 25

Welcome

Welcome

In today's rapidly evolving digital landscape, cloud-native technologies have become a cornerstone for modern application development. Businesses are leveraging the power of cloud platforms to build scalable, flexible, and resilient applications, fostering innovation and accelerating time to market.

Our focus in this module is to provide a holistic approach that integrates the principles of cloud computing, containerization, microservices architecture, DevOps, and continuous delivery. This has given rise to the interdisciplinary field of **Cloud Native Development**, where development and operations converge to create efficient, automated, and agile systems.

Throughout this module, we aim to build your skills in developing, deploying, and managing cloud-native applications. You will gain proficiency in designing scalable applications, optimizing cloud resources, and applying modern cloud technologies to address real-world challenges. Prepare to embark on a journey that will enhance your understanding of cloud-native practices and empower you to lead cloud-driven innovation. Welcome to the future of software development!

Kind regards,

Zeeshan Tariq

Module Coordinator
September 2026

1. Module Overview and Communication

Module Details			
Module Title	Cloud Native Development		
Module Code	COM682	Module Level	6
Credit points	20	Module Status	Core/Compulsory or Optional
Semester	1		
Delivery Mode			
Contact Details and Communication Methods			
Module Coordinator	Zeeshan Tariq (See cover sheet for contact details)		
Teaching Staff Contact Details	Dr. Zeeshan Tariq, z.tariq@ulster.ac.uk Dr. Rashid Kamal, r.kamal@ulster.ac.uk M Shafi (SUST), m.shafi@ulster.ac.uk Teaching Team (QAHE)		
 General Information, Queries and Consultations	If you require advice on any aspect of the module, please read the guidance provided here, in the Module Guide. For General Queries, please utilise the class Discussion Board (if one has been made available in Blackboard Ultra). For Specific Queries or studies advice, you can avail of the 'Drop-in Hours.' Please contact Zeeshan Tariq, if you have questions relating to the module or wish to schedule an appointment. Please note, we aim to respond to emails within 48 hours during the working week.		
 Module Announcements	Key announcements will be presented during tutor-led activities. Out of class communication including notifications, reminders, etc will be distributed via the Blackboard Announcement tool. You will receive a duplication of the announcement direct to your student email inbox. It is essential that you check your emails and Blackboard Ultra regularly for new announcements. Blackboard is mobile responsive and will work on your phone browser, you can also download the 'Blackboard' App as an alternative to access these announcements, notifications and reminders as well as content.		

2. Sustainable Development Goals (SDGs) and Graduate Attributes (GAs) in your module

Sustainable Development Goals (SDGs)

The UN SDGs are a comprehensive set of global goals to end poverty, protect our planet and improve living conditions of the Global Population. You are encouraged to think critically and reflect on the SDGs in the context of this module.

The following SDGs are relevant in the context of this module:

 ☐	 ☐	 ☐	 ☐	 ☐	 ☐
 ☐	 ☒	 ☒	 ☐	 ☒	 ☒
 ☒	 ☐	 ☐	 ☐	 ☐	

Graduate Attributes (GAs)



Ulster University's Graduate Attributes are a high-level set of competencies, developing universal understanding, skills, qualities, and values.

As a result of engaging with this module, you will have opportunities to develop and practise the following selected attributes:

Thriving Individual	D	A
Enhancing Potential	☒	☐
Confidence and Resilience	☐	☐
Well-being	☐	☐
Growth Mindset	☐	☐

Engaged Learner	D	A
Subject Specialist	☒	☐
Creative Problem Solver	☒	☒
Researcher	☐	☒
Critical Thinker	☐	☐

Collaborative Professional	D	A
Responsible Team Player	☐	☐
Effective Communicator	☐	☐
Enterprising Innovator	☐	☐
Digital Fluent Professional	☐	☐

Active Citizen	D	A
Citizen with Integrity	☐	☐
Inclusive Citizen	☐	☐
Sustainable Citizen	☐	☐
Future Thinker	☐	☐

KEY:
D = Developed
A = Assessed

3. Aim and Learning Outcomes



Module Rationale

Contemporary software development and deployment practices have evolved from traditional ones, relying on a variety of novel technologies and concepts. Traditional server infrastructure is supplanted by services offered by dynamic hosting provided by cloud providers such as Amazon Web Services, Google Cloud Platform and Microsoft Azure.

Typical services offered are extensive and archetypally include comprehensive functionality including:

- An overview of cloud native, service development and deployment.
- An introduction to contemporary cloud and Platform-as-a-Service (PaaS) environments, including essential concepts, commands, management, and operational practices.
- Administrative concerns including architectures design, pricing, management, deployment models and related pipelines.
- Strategies to achieve high availability of services.
- Concepts and experience of orchestration of Infrastructure as a Service (IaaS) through scalable cloud providers, including integrating a CI/CD pipeline to these providers.
- Provide students with experience of Platform as a Service (PaaS) concepts.
- Scalable architecture practice incorporating split logic services, client-based workers and distributed caching.
- High-performance service development leveraging advanced techniques, such as serverless execution models, compute services, block and object storage and caching.
- Quantification and profiling of solution performance and costing.

In addition to technical aspects of such modern solutions, there is an impact on the administrative, operational and accounting practices when compared with traditional development and deployment solutions. Through these services, solutions built upon them may react, shrink and grow at scale and on demand whilst reducing overhead and costs. These modern operational practices have shown to increase solution quality with a greater responsiveness to customer needs. This module serves as an introduction to these cloud platforms, related theory, their foundational concepts and operation models.

Module Aim

This module aims to:

- Provide an introduction to contemporary cloud development environments their concepts, commands, management and operation.

- Provide an understanding of the challenges associated with modern operational practices.
- Introducing architectural considerations, pricing and deployment models.
- Provide students with practical and theoretical experience of the scalable, cloud-based, architecture offered by solutions providers such as OpenStack, Amazon Web Services and Azure.
- Provide students with experience of Platform as a Service (PaaS) and Infrastructure as a Service (IaaS) concepts

Learning Outcomes

What you should be able to do by the end of this module?

Successful participants will be able to:

1	Assess the concepts behind a range of cloud native development techniques and critically evaluate when to apply these paradigms to realisation of solutions.
2	Demonstrate a comprehensive understanding of modern cloud development, techniques and practice and how it may be leveraged to address related challenges.
3	Appraise administrative aspects related to cloud native development, such as pricing concerns and access control.
4	Autonomously and independently identify deficiencies when interacting with a range of architectures and deployment paradigms, leveraging knowledge of these deficiencies to improve future practice.

Using Gen AI Tools in this Module

Ulster University encourages the thoughtful and ethical use of Generative Artificial Intelligence (Gen AI) tools to enhance your learning and assessment while maintaining academic integrity. Ethical use means being transparent about when and how AI is used and never submitting AI-generated work as your own. Misuse of AI is considered a breach of Ulster's academic integrity policies.

All use of AI must align with Ulster University's guidance on the responsible and ethical use of AI in education and comply with the specific expectations set out for this module.

It is the responsibility of each student to ensure they are fully aware of the University's requirements regarding the ethical use of AI in their assessments. If you are unsure about what is permitted, always speak with your module coordinator. For full details, refer to the University's Guidance on the Use of AI.

How to Acknowledge the Use of Generative AI

Comprehensive guidance on the use of AI tools in learning and assessment, including how to appropriately acknowledge, reference, or cite Generative AI in assessments, is available at the Guidance on use of AI for students.

4. Assessment and Feedback

What you need to do to demonstrate achievement of learning?

Summary Assessment Information

This module will be assessed via two items of assessment

Type	Assessment method Submission date, time and submission method	Percentage (%) and word count equivalent	Feedback due date
Coursework 1	CW1: Written assignment (50%) Due: Friday 13th November (Noon-Strict deadline) Feedback: Friday 11th December Nature: Online Submission Submit via Blackboard	50%	20 working days post submission date (via Blackboard Ultra)
Coursework 2	CW2: Project (50%) Due: Wednesday 6th January (Noon-Strict deadline) Feedback: Wednesday 3rd February Nature: Online Submission (BBL)	50%	20 working days post submission date (via Blackboard Ultra)

Reasonable Adjustments

Where you have an agreed reasonable adjustment in place, this will be applied to assessments as appropriate. Students are responsible for ensuring that reasonable adjustments are registered with the University in advance of any assessment. If you believe you are entitled to a reasonable adjustment and do not yet have one in place, you should contact Student Wellbeing as soon as possible. Any adjustments will be implemented in line with University policy.

Students should be aware that some programmes have competency standards that students are required to meet, and while every effort will be made to implement reasonable adjustments where possible, these standards apply to all students.

Assessment Guidance

1) Word count includes the title page, contents page, in-text references and citations but excludes tables, reference list and appendices.
2) Your work should include references to relevant journal articles and other good quality information sources and should be properly laid out using the Harvard system of referencing. More detailed guidance is available in the Subject Guides. We encourage you to make good use of all the support services offered by your Campus Librarian, further details are in the Learning Support Services Section.
3) You should refer to the assessment criteria to provide fuller details of the marking criteria for each classification band.
4) In addition, you should refer to the standard assessment guidelines as presented in your Programme Guide/Support Area, this includes guidance and policies on referencing style, plagiarism, etc.
5) Coursework must be submitted by the dates specified. Coursework submitted after the deadline, without prior approval, is not normally accepted. For further guidance on the late submission of coursework, please see the Programme Guide.
6) Assessments must be submitted as per the assessment brief.

   Please note: Words in excess of +10% of the word count will not be marked.

Coursework 1

Assessment Task



Coursework 1 – BRIEF

Weighting	50% of module mark
Submission deadline	Friday noon, 13th November 2026
Feedback due	Friday 11th December (within 20 working days)
Format	PowerPoint slide deck, max 15 slides (+ 1 checklist slide), submitted via Blackboard Ultra
Individual / group	Individual

- Individually design a scalable, cloud-native web application which acts as a multimedia sharing platform, for a use case of your own choosing (e.g. photo sharing, document sharing, audio clips; the media type(s) supported are your choice, informed by what has been covered in labs).
- Your design must be built entirely from Azure resources and patterns covered in Labs 1–6 (see mapping below).

- You are not required to design a generic architecture from first principles. The expectation is that you adapt the pattern demonstrated in Labs (blob storage + document/relational store + Logic App REST API + front end) to your own chosen use case.

Choosing Your Use Case:

Your selected use case must specify:

- Media type(s): chosen from what is covered in labs (images, documents, audio, etc.).
- Target audience: who specifically is this for.
- One driving constraint: something that actually shapes your design decisions (e.g. privacy-sensitive content, high upload volume with large files, geographically distributed users, near-real-time access needs, a retention/compliance requirement).

Example: not "a website where users share photos," but "X-ray image sharing for rural GP practices referring cases to a regional radiologist, where upload volume is low, but files are large and access must be tightly restricted." Same underlying Azure pattern, entirely different justification and scalability discussion, which is exactly what is assessed.

Pick any domain below or propose your own if it is not on the list. Multiple students choosing the same domain is fine, but your three differentiators (media type, audience, constraint) should still make your design distinct from a classmate's.

1. Medical imaging sharing for remote/rural clinics	16. E-learning lecture recording sharing
2. Citizen-science wildlife photo/video sharing	17. Veterinary clinic pet health image sharing
3. University coursework video submission portal	18. Construction site progress photo sharing
4. Sports team highlight clip sharing	19. Disaster/emergency response photo reporting
5. Security/CCTV footage sharing for a small business	20. Agricultural drone imagery sharing
6. Podcast episode/clip sharing	21. Fashion/design portfolio sharing
7. Recipe video sharing for a cooking community	22. Genealogy/family archive photo sharing
8. Event photography sharing (weddings/conferences)	23. Research dataset/image sharing (non-medical)
9. Insurance claim photo evidence upload	24. Neighbourhood-watch photo/video reporting
10. Legal document/exhibit sharing for a law firm	25. Restaurant menu/food photo sharing
11. Real estate property photo/video listings	26. Fitness progress photo/video sharing
12. Music demo sharing for independent artists	27. Travel blog photo/video sharing
13. Field engineering site inspection reports	28. Open-source project screenshot/doc sharing
14. Citizen journalism/eyewitness photo submission	29. Local council public-works photo reporting
15. Museum/gallery digital exhibit sharing	30. Propose your own (state audience + constraint clearly)

Design Requirements:

Every design element below should be traceable to something you have already done hands-on by the time you submit. This is deliberate: if you can build it in a lab, you know it works, and you can design it with confidence.

1. Problem & scalability discussion

   Discuss the technical problem behind your chosen use case and the scalability issues that a traditional (non-cloud) solution would face for the resources you plan to use.

   Underpinned by: Lab 1 (cloud computing concepts, cost/scale rationale)

2. Webpage wireframe

   Design a wireframe / layout diagram for your multimedia hosting webpage.

   Underpinned by: Lab 4-b (Figma wireframing)

3. Solution architecture diagram

   Present a network/architecture diagram using only Azure resources you have used in Labs 1–6: static website hosting, blob storage, SQL and/or Cosmos DB, and a Logic App REST API layer.

   Underpinned by: Labs 1, 4-a, 6

4. Database schema design

   Design the schema needed to store user entries (e.g. image/file metadata: id, fileName, userID, userName, fileLocator, filePath, or your own equivalent set).

   Relational option: present as a normalised (3NF) table list and/or ERD.

   Non-relational option: present as commented JSON or an equivalent document/table layout.

   Underpinned by: Lab 2 (Azure SQL, 3NF, ERD), Lab 3 (Cosmos DB documents), Lab 6 (document schema for image metadata)

5. REST API design

   Design REST endpoint(s) to support create, retrieve, update and delete operations for your asset records, hosted via Logic Apps. Present this as a UML diagram.

   Model this directly on the IUPS (upload) and RAI (retrieve-all) endpoint pattern demonstrated in Lab 6 — extend it to cover update and delete for your use case.

   Underpinned by: Lab 5 (REST concepts, Postman), Lab 6 (Logic App REST endpoints, CRUD pattern)

6. Integration overview

   Show how a statically hosted front end, REST endpoints and your chosen database(s) connect together — i.e. the same shape as the Lab 6 solution, applied to your use case.

   Underpinned by: Lab 4-a (static website hosting), Lab 6 (end-to-end wiring)

7. Assessment of limitations

   Discuss the limitations of your designed solution and, where possible, how they might be addressed.

8. Assessment of scalability

   Assess how your solution design supports scalability (e.g. horizontal/vertical scaling, autoscaling, serverless consumption models as seen in labs).

9. Concluding comments

   Reflect on how the solution could be improved, its commercial usefulness, and any ethical considerations.

10. References

Harvard-referenced sources, frameworks, or Azure documentation used to inform your design.

Submission Format

- Slide deck produced in PowerPoint format, uploaded to the relevant Blackboard assessment area.
- Maximum 15 slides. Slide notes and any slides after slide 15 will not be assessed.
- A submission checklist slide (see below) must be included at the end and is not counted within the 15-slide limit.

Content Outline for Slides

- Title slide: project name, one-line description of your use case, your name and student number.
- Discussion of the technical problem and scalability issues affecting the resources used in your project (not a description of project objectives).
- Solution architecture of the project, using only Azure resources covered in Labs 1–6.
- Database schema design (relational and/or non-relational).
- REST API design (UML), modelled on the Lab 6 CRUD pattern.
- Assessment of the limitations of the solution.
- Assessment of scalability of the designed solution.
- Concluding comments — improvements, design limitations, commercial usefulness, ethical considerations.
- References.

Submission Checklist

Before submitting, make sure you have addressed the following:

General

- Did you include a title slide with project name, one-line description of your use case, your name, and student number?
- Is your submission in PowerPoint format and within 15 slides (excluding the checklist slide)?

Problem & Context

- Did you discuss the problem related to your use case, indicating why a cloud solution is appropriate?
- Did you specify a target audience and a driving constraint for your use case, beyond restating “a multimedia sharing platform”?
- Did you identify scalability issues affecting the resources in your project?

Solution Design (Labs 1–6 only)

- Did you include a solution architecture diagram using only Azure resources covered in Labs 1–6?
- Did you design and present a multimedia hosting webpage layout (wireframe)?
- Did you design a database schema (SQL 3NF/ERD and/or NoSQL document layout) for storing user entries, consistent with the Lab 6 pattern?
- Did you design a REST API for CRUD operations, presented as a UML diagram, modelled on the Lab 6 IUPS/RAI endpoints?
- Did you show how the static front end, REST endpoints and database(s) connect together?

Critical Evaluation

- Did you include an assessment of the limitations of your designed solution?
- Did you include an assessment of scalability (horizontal/vertical scaling, autoscaling, etc.)?
- Did you justify why you selected specific Azure services over alternatives?

Conclusion & References

- Did you include concluding comments summarising your design?
- Did you provide references to sources, frameworks, or Azure documentation used?

If you can tick all these boxes, your submission is likely to meet expectations. Please include this checklist as the final slide of your deck (it is not counted in the 15-slide limit and not graded).

Coursework 1 – ASSESSMENT RUBRIC/MARKING PROFORMA

Assessment criteria	80+ 1st	70-79 1st	60-69 2:1	50-59 2:2	40-49 3rd	<40 Fail
Problem Definition & Discussion (20%)	As for 70-79, with an excellent, comprehensive justification of the problem and solution rationale, of a standard that could be shared as an exemplar or used to teach others; an in-depth, publishable-quality analysis of how cloud computing, associated patterns and architectural components are being used.	A good, integrated description of the problem with clear justification of why a cloud solution ought to be developed; strong critical appraisal of cloud technologies, related patterns and architectural components, synthesising how they combine to address the problem, with insight into how the approach could transfer to related contexts.	A sufficient, connected justification of why a cloud-based solution needs to be developed, with a good description of the problem as a whole; appropriate critical evaluation of how cloud technologies and related patterns interact and are being used together in the solution.	Adequate description of the problem with moderate justification for a cloud solution; an adequate critical appraisal of the use of cloud technologies and related patterns, considered individually.	Insufficient explanation of the issue as a whole; inadequate justification for a cloud-based model; a constrained, selective analysis of cloud technologies used in isolation.	Little or no description of the overall problem; poor or absent justification for why a cloud-based solution is needed; little to no critical appraisal of the use of cloud technologies.
Overview of the Technical Solution Developed (40%)	As for 70-79, with a thorough, in-depth investigation, analysis and selection of technology, presented to a publishable standard; alternative technologies investigated and detailed with pros/cons; a wide range of cloud-	The technology used was carefully examined and logically chosen, with alternative technologies investigated and reasons for exclusion given; a wide range of cloud-native components are incorporated and shown	Given the development issue, the technology used is logically chosen with sufficient supporting detail; cloud-native components are incorporated with appropriate effort and interact coherently, informed by cloud-	The technology used is appropriate given the development problem, with sufficient detail provided in isolation; moderate effort to incorporate cloud-native components, informed by cloud-native design patterns considered	Minimal justification for the technology choice; design poorly informed and incorporates few cloud-native elements; no meaningful solution architecture presented.	No justification for the choice of technology applied; no meaningful design presented; no cloud-native elements incorporated.

Assessment criteria	80+ 1st	70-79 1st	60-69 2:1	50-59 2:2	40-49 3rd	<40 Fail
	native components incorporated with insight into their advantages over other cloud-based solutions in the literature, of a quality that could be used to teach others.	to interact and depend on one another; a well-documented, justified architecture incorporating control flows and software architecture diagrams demonstrates how the design could be adapted to related solutions.	native design patterns; a detailed architecture diagram is presented showing how components connect.	individually; an architecture diagram of the developed solution is presented.		
Assessment of Limitations (15%)	As for 70-79, with a comprehensive, publishable-standard analysis of the constraints imposed by the solution and potential solutions to these problems, of a quality that could be used to teach others.	A broad, connected appraisal of the limitations of the solution, with strategies to address them and insight into how addressing one limitation affects others.	Limitations discussed with awareness of how to remedy them, and how these limitations relate to and affect one another.	Some limitations discussed with some awareness of how to remedy them, considered individually.	Limitations discussed without any reflection.	Limitations of the solution not enumerated nor discussed adequately.
Assessment of Scalability (15%)	As for 70-79, with scalability effectively and comprehensively addressed by employing a number of cloud-native components throughout the development process, to	Scalability was well catered for with multiple cloud-native elements applied, and their combined, synergistic effect on scalability is clearly explained.	Scalability was catered for across most of the resources in the solution, with multiple cloud-native elements applied and shown to work together.	Scalability was partially catered for in the solution, with several relevant elements present but not clearly connected.	The solution minimally integrates elements to offer scalable operation, considered in isolation.	The solution integrates no elements to offer scalable operation.

Assessment criteria	80+ 1st	70-79 1st	60-69 2:1	50-59 2:2	40-49 3rd	<40 Fail
	a standard that could be used to teach others.					
Concluding Comments (5%)	As for 70-79, with the solution, its functionality, limitations and potential applicability given careful consideration in an in-depth, publishable-quality analysis, of a standard that could be used to teach others.	Insightful reflection was applied to the solution, its functionality, limitations and potential applicability, synthesising what the problems were and proposing potential, transferable solutions.	A variety of meaningful, connected reflection was applied to the solution, its functionality, limitations and potential applicability; weaknesses were identified and improvements suggested.	Meaningful reflection was applied to the solution, its functionality, limitations and potential applicability, considered individually.	The proposed solution, its functionality, limitations and potential applicability were given some reflection, in isolation.	No reflection was applied to the solution, its functionality, limitations or potential applicability.
Referencing (5%)	As for 70-79, with referencing and use of evidence of a professional, publishable standard.	Correct and appropriate referencing throughout, with accurate in-text citation and a consistent, correctly formatted reference list.	Relatable, appropriate references used with mostly consistent application of the referencing convention.	A limited range of references used, with inconsistent application of the referencing convention.	Inadequate or incorrect referencing; sources included are not clearly related to the work.	No referencing.

Coursework 2

Assessment Task



Coursework 2 – BRIEF

Weighting	50% of module mark
Released	Week 1
Submission deadline	Wednesday noon, 6th January 2027
Feedback due	Wednesday 3rd February 2027 (within 20 working days)
Format	5-minute maximum video walkthrough (Panopto Capture preferred), submitted via Blackboard Ultra
Individual / group	Individual

- Implement, deploy and test the solution you designed in CW1, using the Microsoft Azure cloud platform and the practical techniques covered in this module. If your implementation is different than the design presented in CW1, please clearly mention and justify in the CW2 submission.
- The developed solution should implement full CRUD (create, retrieve-all, retrieve-individual, update, delete) for your asset records, regardless of whether your metadata store is Azure SQL Database or Cosmos DB.

Implementation Requirement

1. Core implementation, based on your CW1 design
   - Implement the solution based on the design produced in CW1.

   Underpinned by your CW1 design + Labs
2. Scalable blob storage
   - Use scalable cloud-native storage (Azure Blob) for binary data such as uploaded files.

   Underpinned by: Lectures, Lab 4-a, Lab 6 (2.1 Establishing a blob store)
3. Metadata database
   - Leverage a cloud-native database to store metadata for your asset records — either Azure SQL Database or Cosmos DB, consistent with your CW1 design.

   Underpinned by: Lab 2 (Azure SQL), Lab 3 (Cosmos DB), Lab 6 (2.2 Establishing a Document Store), Lab 8.
4. REST API — full CRUD
   - Design and implement REST endpoints supporting create, retrieve-all, retrieve-individual, update and delete for your asset records, hosted via Logic Apps.

    - SQL-based solutions: follow the RAA/CIA/RIA/UIA/DIA pattern and URI/verb mapping table from Lab 8.
    - Cosmos DB-based solutions: follow the IUPS/RAI pattern from Lab 6, extended with the RIPS/UIPS/DIPS endpoints — same Cosmos DB connector, same Logic Apps designer, additional Get document / Delete document actions mirroring what Lab 8 does for SQL.
- Create endpoint URIs that store and retrieve images/files and their metadata.

Underpinned by: Lectures, Lab 5 (REST/Postman), Lab 6 (2.3 Creating a REST API, extended), Lab 8 (2.3 REST API Implementation), Lab 9 (sample apps for both patterns)

5. CI/CD build pipeline
    - Set up a CI/CD build pipeline for your front-end using Azure DevOps Pipelines connected to a GitHub repository, following the Lab 10 pattern (push your code via git, connect the repo to a pipeline, and produce a working build).
    - Automatic deployment of that pipeline to a live host is a bonus if you achieve it but is not required for full marks — Lab 10 only covers this as optional further reading, not a guided hands-on step, so it is not fair to require it outright.

Underpinned by: Lectures, Lab 10 (Create a CI/CD Pipeline on Azure)

6. Advanced Azure features
    - Add least two advanced features to your solution, clearly identified and demonstrated working in your video. Options actually taught hands-on in this module are following, but not limited to:
        - Azure Functions: an HTTP-triggered function (Python or Node.js) supporting your solution (Lab 7).
        - Custom Question Answering (QnA), e.g. an FAQ/help feature for your platform, built from your own content (Lab 11).
        - Azure AI Translator, e.g. translating captions, descriptions or metadata text (Lab 12).

If you choose not to add advanced features, or only partially implement, clearly state this in your video. You will still receive marks for the rest of a working solution, but this affects the Advanced Features criterion specifically.

Underpinned by: Lectures, Lab 7, Lab 11, Lab 12

Submission: Video Walkthrough

Each student individually submits a single video, 5 minutes maximum, walking through the developed, tested and deployed solution.

Use the video to showcase functionality and deployment to Azure — do not re-run through your CW1 slides.

If there is no valid excuse, include yourself on camera (turn your camera on if possible).

Record using Panopto Capture (preferred) via the 'Panopto' link on the Blackboard menu, or an alternative dual-recording tool such as screencast-o-matic.com.

Content Requirements for the Video

Your running application, performing full CRUD via RESTful APIs. Create, retrieve, update and delete should all be working for maximum marks — partial marks are available for a partially completed solution.

Your backend Azure resources: Logic Apps, database(s), blob storage, and your CI/CD pipeline.

URIs referring to the Azure resources used for your API operations.

Clearly identify which advanced feature(s) you have used and demonstrate them working — or clearly state if none were added.

If your implementation differs from your CW1 design in any way, briefly explain why.

Plagiarism

N.B. Students should be aware of the plagiarism policy of the University and submit their coursework in accordance with this. Plagiarism is the unattributed copying of the work of another person, either from a published work or the work of another student. It is a form of literary theft and is not permitted under any circumstances. Plagiarism is regarded by the University as a very serious offence and subject to formal disciplinary proceedings. See [3] for more details.

I declare that this is all my own work. Any material I have referred to has been accurately referenced and any contribution of Artificial Intelligence technology has been fully acknowledged. I have read the University's policy on academic misconduct and understand the different forms of academic misconduct. If it is shown that material has been falsified, plagiarised, or I have otherwise attempted to obtain an unfair advantage for myself or others, I understand that I may face sanctions in accordance with the policies and procedures of the University. A mark of zero may be awarded and the reason for that mark will be recorded on my file.

Penalties for exceeding the assessment limits

Where submitted work exceeds the agreed assessment limit, a margin of up to +10% of the work limit will be allowed without any penalty of mark deduction.

If the work submitted is significantly in excess of the specified limit (+10%), there is no expectation that staff will assess the piece beyond the limit or provide feedback on work beyond this point. Markers will indicate the point at which the limit is reached and where they have stopped marking.

A mark will be awarded only for the content submitted up to this point. No additional deduction or penalty will be applied to the overall mark awarded. The student is self-penalising as work will not be considered/marked.

Disciplinary procedures due to Academic Misconduct may be invoked if the word count/workload has been deliberately and significantly falsified.

Coursework 2 – ASSESSMENT RUBRIC/MARKING PROFORMA

Assessment criteria	80+ 1st	70-79 1st	60-69 2:1	50-59 2:2	40-49 3rd	<40 Fail
Implementation (35%)	As for 70-79, with an exceptional, innovative implementation with all components expertly integrated and functional, of a standard that could be used to teach others.	Detailed and highly effective implementation with all components seamlessly integrated and functional, demonstrating a synthesised, coherent solution that could be adapted to related use cases.	Comprehensive implementation with all major components functional and well-integrated, working together coherently.	Adequate implementation with most components functional, though minor errors may exist; components not always shown to work together.	Basic implementation with significant issues or missing components, considered in isolation.	Incomplete or incorrect implementation; key components are missing, non-functional, or improperly deployed.
Use of Azure Resources (35%)	As for 70-79, with masterful use of Azure resources, flawless deployment and integration, of a standard that could be used to teach others.	Excellent, synthesised use of Azure resources with all components flawlessly deployed and integrated, demonstrating understanding of how resources combine and could transfer to related solutions.	Effective use of Azure resources with all components correctly deployed and well-integrated, interacting appropriately.	Adequate use of Azure resources with most components correctly deployed and functional.	Some Azure resources used, but implementation is flawed or incomplete, with resources considered in isolation.	Poor or incorrect use of Azure resources; key services missing or non-functional.
Use of Advanced Features (20%)	As for 70-79, with advanced features integrated at an expert level, adding significant value, of a standard that could be used to teach others.	Multiple advanced features expertly integrated and fully functional, significantly enhancing the solution and demonstrating synthesis with the core implementation.	Effective integration of multiple advanced features with good functionality, working coherently with the rest of the solution.	Some integration of advanced features, with basic functionality.	Minimal integration of advanced features, with limited functionality, considered in isolation.	Little to no attempt to integrate advanced features; features non-functional or incorrectly implemented.
Video Quality and Presentation (10%)	As for 70-79, with an exemplary video of a highly professional, publishable standard, clearly structured, that could be used to teach others.	Highly professional video, clearly structured, presenting a coherent narrative of the solution, within the time limit.	Well-structured and clear video, presenting the solution effectively, within the time limit.	Clear video that presents the solution adequately but may lack depth.	Somewhat clear video but may be poorly structured or lack essential content.	Poorly structured video, unclear, lacks key elements; student may not appear in video.

5. Learning Resources

A list of current learning resources specifically chosen to build your knowledge and understanding for this module

Reading List - Also available online via Key Links:

Books/Journal Article/Publications Required Reading (Must read)

Microsoft Corporation, (2017) Cloud Application Architecture Guide, Microsoft press. Seattle.

Azure Architecture Centre (2024) , Microsoft, https://learn.microsoft.com/en-us/azure/architecture/

Books/Journal Article/Publications Recommended Reading (Should read/Could read)

Scholl, B., Swanson, T., & Jausovec, P. (2019). Cloud Native: Using Containers, Functions, and Data to Build Next-Generation Applications. O'Reilly Media.

Caldato, C., (2020) Cloud Native for the Enterprise, O'Reilly Media, Inc. Sebastopol.

Gilbert, J., (2018) Cloud Native Development Patterns and Best Practices: Practical architectural patterns for building modern, distributed cloud-native systems, Packt Publishing. Birmingham.

Learning Microsoft Azure: Cloud Computing and Development Fundamentals (2023), O'Reilly Media, Inc. Sebastopol.

Useful Journals

Useful Library Databases and Websites



Library's Support Services

The Library is here to support you throughout your academic journey; at the start of the year, you will be invited to introductory presentations and guided tours to help you get familiar with our space's services and resources. The Academic Engagement team works closely with Faculty to deliver specialist classes and workshops to help you effectively utilise Library services and information resources to complete your assignments.

Visit the Library Homepage to access contact details, Subject Guides, referencing support and the Training & Workshop calendar.
Whether you need guidance on referencing, assignment planning or improving your research skills, the library is here to help!

Blackboard Learning Support

The Student Blackboard Support Wiki provides advice and how-to guides to help you use Blackboard Learn effectively.

If you require help or support with any of the digital learning tools utilised within your module, please contact The Blackboard Helpdesk via:

028 9536 7188, or by e-mail

IT Support

Digital Services provides and maintains the University's core ICT services, supporting students with the secure setup of their email account, connecting to Wi-Fi, and accessing devices and software in teaching spaces and the Library.

IT Resources:

To help you get started, Digital Services have created a **Getting Started IT Guide**, with information on how to access your account, student email, network password and other resources to help you.

You should also visit our dedicated Student resources pages to find information on the technologies, software and support provided to our students.

How to Get Help:

The IT Service Desk Team is here to help you make the most of your studies by providing support with:

- Accessing Microsoft 365 (student email, OneDrive), Student Portal, and Blackboard
- Setting up multi-factor authentication, password resets, and downloading student software
- Connecting to Eduroam Wi-Fi, printers, and troubleshooting IT issues

How to contact IT Service Desk:

- Email: Contact servicedesk@ulster.ac.uk
- Phone: Call 028 9536 7776

- Online: UniDesk Self-Service Portal:
- On Campus: Drop-in IT support in our libraries

6. Organisation and Management

Types of learning activities that will make up your weekly timetable



COM682 Cloud Native Development is a 20-credit point module, this requires approximately 200 hours of your commitment, distributed through the following learning and teaching activities over the 15-week semester. For a description of the nature of the learning activities please refer to your Programme guide.

Summary of Learning Activities

Learning Activities: Week 1-12	Indicative Weekly Hours	Total Hours (200 hours)
Lecture and Class Activities	2 hours	24
Practical Activities in Lab environment	2 hour	24
Independent Study: Week 1-15	**Indicative Guide**	
Independent Study	10 hours approx. weekly	152

The teaching and learning plan provides a more detailed overview of content on a weekly basis.

Module delivery will typically consist of a two-hour lecture and a one-hour Seminar although format may vary slightly between weeks/sessions.

Class Timetable

For the day, time and room number, view your timetable on PUBLISH.

Your timetable is shared with you on Publish. You can access it from the student portal by clicking the Class Timetables tile. It lists the classes you should attend, such as lectures, seminars, workshops, etc. Support is available on the Timetabling webpage.

Seminars and Attendance

Please refer to the Teaching and Learning Plan below for details of seminar topics. Attendance at seminars is essential and will be monitored.

All students are expected to download the SEAtS Student Attendance App via the App Store or Google Play to register their attendance when they arrive for each class. Regular check-ins using the app help you stay on track with your studies and maintain an accurate attendance record.

You can find everything you need to know about checking into your classes—including user guides, video demonstrations, and FAQs—on the SEAtS attendance information page.

Teaching and Learning Plan

The teaching plan/order of weekly topics is subject to change. The list below provides an outline summary of weekly activities and further information is available on Blackboard Ultra in the relevant week's folder. Students will be expected to engage in all prescribed activities that contribute towards final assessment.

Week/L&T Methods	Curriculum Content/Topic Title	Independent/Directed Study, includes: Required Reading, Seminar Preparation, and/or Assessment Preparation and due dates
Week-1	Module Intro & history	Lab-01 Creation of Azure Accounts
Week-2	Cloud Computing Concepts & features overview	Lab-02 Azure SQL
Week-3	No Sql and Document Databases	Lab-03 Cosmos DB
	Hosted Relational databases	Lab-04 Azure Storage Services; Wireframing (Figma)
Week-4	Cloud Design Patterns and Architecture I & II	Lab-05 Azure REST APIs
Week-5	Hosted Web Endpoints (REST)	Lab-06 Designing Storage Databases and Creating REST APIs
	Azure Data storage Platforms	Lab-07 Multi-tenant Logic Apps; Azure Functions (Python & Portal)

Week/L&T Methods	Curriculum Content/Topic Title	Independent/Directed Study, includes: Required Reading, Seminar Preparation, and/or Assessment Preparation and due dates
Week-6	Decomposing a Cloud: Virtualization, Containerization and Serverless Execution	Lab-08 Creation of Logic Apps and REST Endpoint (full CRUD via Azure SQL)
Week-7	Caching, Content Delivery Network	Lab-09 Sample Web Apps for Asset Tracking (Lab 8) & Multimedia Sharing (Lab 6)
	Serverless Computing on Azure	Lab-10 CI/CD Pipeline (Azure DevOps + GitHub)
Week-8	Azure Networking, Azure DNS, and Routing	Lab-11 Custom Question Answering (QnA) Project
	Load Balancing and Scalability	Lab-12 Azure AI Translator REST APIs
Week-9	Azure Monitor & Azure Application Insights Profiler	Lab-01 Creation of Azure Accounts
Week-10	Open Source Licenses, CI/CD	Lab-02 Azure SQL
Week-11	**Advanced Azure Services –Cognition II**	Lab-03 Cosmos DB
Week-12	**Advanced Azure Services –Cognition II**	Lab-04 Azure Storage Services; Wireframing (Figma)

7. Student Voice and Support Services

How we support you



As a course team, we incorporate the key partnership principles set out in the joint UU & Ulster University Students' Union Student Voice Guidelines and proactively engage with the democratic election of UUSU academic student reps (Faculty Reps, School/Dept Reps & Programme Reps) to ensure that student opinion is heard at Ulster. We respect your views and welcome your honest and constructive feedback on the module.

There are several ways to do this:

You can contact your Module Coordinator about any queries related to your learning experiences on the module as/when you have them.

You can voice your opinions through the formal Staff/Student Consultative Committee process by contacting one of the elected UUSU Programme Reps in your class.

You will have the opportunity to give feedback on the module through completing the online Student Module Feedback Survey.

UUSU Advice Bureau Service

If you are experiencing difficulties that are impacting your studies, you can contact the Advice Bureau in the Students' Union (UUSU). You can get advice and guidance on issues such as complaints, appeals, housing problems, disciplinaries, and information on various support providers available. To have a chat with the team, contact UUSU online.

UU Student Wellbeing Service

Ulster University's Student Wellbeing team is here to help you manage a range of common challenges you may face during your studies. These include stress, academic or relationship issues, and financial or budgeting challenges.

The team also provides support for students who require reasonable academic adjustments due to a disability or specific learning difficulty, such as dyslexia.

There is no stigma in seeking help. Accessing the right support can make a real difference to your wellbeing and academic success at Ulster University.

UU Student Success Centre

Ulster University's Student Success team has developed a series of academic and study skills training resources and workshops to help you succeed educationally and develop personally and professionally. You can access these resources via the Student Success website or you can contact the team directly via email. You can also request support by completing the Online Referral Form, available via the Student Success Centre tile on the Student Portal.